Local effect of AGN on the host galaxy

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Abstract

The role of Active Galactic Nuclei on the star formation of their host galaxies is an intensely debated issue nowadays. Here we use CALIFA and AMUSING data to examine the evolution of the Star Formation Histories in search for these effects, distinguishing whether we take into account the whole galaxy or only its circumnuclear region and classifying our sample of galaxies in terms of their nuclei, morphology and mass.

Madrid, July 8, 2018

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I. Introduction

The main processes that are ultimately responsible for regulating star formation in galaxies remain unknown. Several studies[1;2;3] have shown a bimodal distribution of galaxy colours, indicative of the ratio between young (recently formed) and old stars, that is often interpreted in terms of two distinct populations of "star-forming" and "passive" galaxies. The colour of a galaxy is also strongly correlated with its overall mass and morphology. While massive, early-type (spheroidal) systems are generally redder, low-mass, late-type (disk or irregular) galaxies tend to display blue colours, consistent with a roughly constant star formation rate (SFR, the total mass of stars formed per year). In between these two populations, usually referred to as the "red sequence" and the "blue cloud" respectively, an underpopulated zone of the colour-magnitude diagram called the "green valley" is viewed as a crossroads in galactic evolution, representing a transition period between the blue cloud of star-forming galaxies and the red sequence of passively evolving systems[4;5;6;7;8;9;10].

Many of the galaxies in the "green valley" show signatures of an Active Galactic Nucleus (AGN) at their centre[11;12;13], i.e. energetic radiation that cannot be attributed clearly or directly to the stars. The first optical spectrum of one of these objects was obtained by E. A. Fath in 1909[14], who observed the intense emission lines in NGC1068, and it was four decades later when Carl Seyfert[15] realized that there were several similar galaxies of such kind. It is nowadays assumed that AGNs are powered by the accretion of matter onto a Super-Massive Black Hole (SMBH), with almost 10% of the accreted energy being radiated away as light[16]. If the AGN is bright enough, the pressure exerted by the released photons can prevent further infall of matter towards the black hole, regulating its growth. If the luminosity rises above the Eddington Limit*, the surrounding gas may actually be ejected out of the disk. The correlation between the estimated properties of the central SMBH and those of the host galaxy[17;18;19] together with the observation of high-speed winds in the immediate vicinity of some AGN[20;21] have led to the hypothesis that AGN-driven outflows may blow away a significant fraction of gas reservoir, rapidly "quenching" (producing a sudden decrease that anihilate the star formation rate) both black hole growth and star formation within the galaxy[22;23;24;25;26].

Nevertheless, causal connection is far from proven. Different analyses[27;28] of the optical spectra of a large sample of nearby galaxies cast serious doubts on the idea that there is a sharp transition between star-forming and passive systems, suggesting instead that galaxies are distributed along a gradual "ageing" sequence. Previous work carried out by another student[29] based on a sample of ~ 300 galaxies from the Calar Alto Legacy Integral Field Area (CALIFA) survey[32;33;34] did not

*__Eddington Limit:__ The maximum luminosity a body can achieve when there is balance (hydrostatic equilibrium) between the outward radiation pressure and the inward gravitational force.

find any evidence of quenching in any of the galaxies nor any statistically-significant correlation between the star formation history (SFH, how stars are formed over time and space) and the presence or absence of AGN signatures in the optical spectrum.

The present study builds upon those results, expanding the galaxy sample (with additional observations from the CALIFA survey as well as newer data with higher spatial resolution from AMUSING) and focusing on the possible effects of the AGN not only at the scale of the whole galaxy but also in the central regions, where they are expected to be more evident. The observational data and their analysis procedure are described in Section II and III. The main results are presented in Section IV, and Section V summarises our main conclusions.

II. Observations

1. Galaxy sample

Our analysis is entirely based on integral-field spectroscopy (IFS) data. This observational technique allows the simultaneous measurement of a "datacube" containing both spatial and spectral information; each datacube can be viewed as a collection of "slices" (images of the object at different wavelengths) or "spaxels" (individual spectra at a given spatial location). The number of IFS instruments has steadily increased over the last years, and the number of available observations has grown from a handful of isolated objects to systematic studies of statistically-significant samples (e.g. MaNGA[30], SAMI[31]...).

The vast majority (584) of the galaxies considered in the present work have been obtained from the Third Data Release (DR3) of the Calar Alto Legacy Integral Field Area (CALIFA) survey[32;33;34]†. Such observations were carried out with PMAS/PPAK integral-field spectrophotometer, mounted on the 3.5-m telescope at Calar Alto Observatory, Spain. They cover a wavelength range from 3700 to 7500 Å in two overlapping setups, a high-resolution mode (V1200, $3700 - 4200$ Å) and a low-resolution mode (V500, $3750 - 7500$ Å), allowing the measurement of the most important spectral features in the optical range. The datacubes are sampled with $1'' \times 1''$-spaxels over a hexagonal field of view (FoV) of $\sim 74'' \times 64''$. The point-spread function (PSF, indicative of the spatial resolution) has a typical full-width at half-maximum (FWHM) of the order of $2.7''$, and the survey has selected its targets from the photometric catalogue of the Sloan Digital Sky Survey (SDSS) DR7 according to their angular diameter on the sky, adopting a combination of isophotal diameter‡ and distance (redshifts $0.005 < z < 0.03$) criteria, in order to ensure an optimal coverage

†`http://www.caha.es/CALIFA/`

‡**Isophotal diameter:** Diameter at which a particular surface brightness level is reached.

of the whole extent of the galaxy (at least two effective radii[§]) by the PPAK instrument.

In addition, we complement our galaxy sample with 90 objects from the All-weather MUse Supernova Integral-field of Nearby Galaxies (AMUSING) survey. These data have been obtained with the Multi-Unit Spectroscopic Explorer (MUSE) instrument installed at the UT4 telescope of the Very Large Telescope (VLT) of the European Southern Observatory (ESO) in Cerro Paranal, Chile. The FoV is ~ 1 arcmin^2, sampled with $0.2'' \times 0.2''$ spaxels (the final science data contain $320 \times 322 \sim 10^5$ spectra per galaxy!). The actual spatial resolution is limited by the excellent quality of the atmospheric seeing at Cerro Paranal, which is usually characterised by a PSF of the order of $\sim 1''$ FWHM. These data provide an improved view of the circumnuclear region, which enables a much more accurate separation of the stellar and AGN contribution.

2. Resolved measurements

For each galaxy, we use local measurements of the star formation history (SFH) and the intensity of nebular emission lines carried out with Pipe3D[36], a software package specialised in the analysis of IFS data that is based on the Fit3D tool[35] and developed to decompose each individual spectra as a sum of simple stellar populations (SSPs) with different ages, metallicities, and kinematics, as well as emission from the warm ionized gas.

Pipe3D takes the measured flux densities and the estimated errors in the datacube as input and returns a large number of physical properties as output. The algorithm consists in a number of steps that we briefly summarise here:

1. Extract a central spectrum of each data cube ($2.5''$ radius) and fit it using a reduced set of SSP templates plus a typical spectrum of an emission line source, with the objective of deriving the systemic velocity of the galaxy and the central velocity dispersion.

2. Group contiguous spaxels into spatial bins until a minimum signal-to-noise ratio (S/N) is reached. The signal in the central parts is usually sufficient to keep individual spaxels separate, but the surface brightness declines rapidly with galactocentric distance, and large bins are necessary to carry out meaningful measurements in the outer regions. The output of the binning scheme is stored as a segmentation map with the label of the spatial bin that each pixel belongs to.

3. For each spatial bin, the summed spectrum is analysed by Fit3D using a multi-SSP decomposition technique[42] based on a database of spectral templates that combines synthetic stellar

[§]**Effective radius:** The radius such that half of the galaxy light is emitted from within.

spectra from the GRANADA library[43] and empirical spectra from the MILES project[44;45;46], assuming a Salpeter (1955)[47] power-low as Initial Mass Function (IMF, the fractional distribution of mass of a newly formed population of stars). This procedure returns the fraction of light α_{SSP} contributed by each SSP, from which the main properties of the stellar populations (such as mean age or metallicity, full SFH, etc.) within each bin can be derived.

4. In order to make detailed measurements in individual spaxels, a "stellar continuum" datacube is generated by rescaling the best-fitting SSP model according to the local flux intensity observed in each spaxel. The difference between the original and stellar-continuum datacubes is attributed to nebular emission, and the intensity I_λ of the different emission lines is estimated by fitting a Gaussian function.

III. Analysis

1. Regions of interest

Our analysis starts from the resolved maps of the SSP coefficients α_{SSP} and line intensities I_λ returned by Pipe3D for each of our galaxies. We will integrate the single-spaxel measurements over three different regions of interest:

1. In order to characterise the nuclear activity of all our galaxies, we will consider the intensity of several emission lines (see below) within a circular aperture ensuring that the fractional contribution of all the spaxels within the exact circular aperture is properly taken into account. In the case of CALIFA data, this aperture was taken to be a $1''$ radius, and for AMUSING data it was 3 pixels.

2. We define the circumnuclear region, where the impact of an AGN is potentially more obvious, as an annulus with a fixed physical size of 1 kpc at the outer radius[¶] for both samples, and an inner radius of the respectively aforementioned nuclear size in order to minimise contamination from its emission. (See Figure 1)

3. Finally, we also consider the global star formation history of the entire galaxy for CALIFA sample (the AMUSING field of view does not allow this analysis as it only contains parts of the galaxies). Emission-line ratios integrated over the whole field of view are also investigated in order to assess aperture effects that would apply to objects at larger redshifts.

[¶] 13 of the objects in our CALIFA sample are located further than 200 Mpc, and therefore 1 kpc subtends less than $1''$ projected on the sky. To have the sufficient annular-area to be analysed, it was imposed that 1kpc>1.5". Galaxies that do not match this restriction were eliminated from the sample.

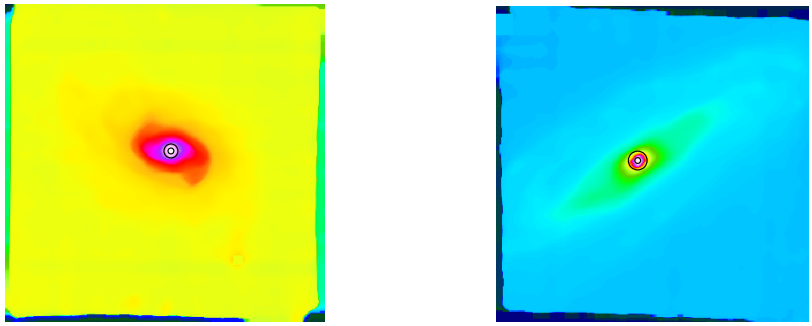

Figure 1: Annular region for two objects in AMUSING sample (left: 2MASXJ01504127-1431032, right: NGC5185). The outer radii correspond to 1kpc and inner to 3 pixels.

2. Galaxy classification

We have classified the galaxies in our sample into different groups according to their mass, optical morphology, and nucleus activity.

The total stellar mass of each object has been computed by adding the mass surface density in all spaxels and then multiplying by their corresponding physical area $a = \Delta\Omega\ D^2$, where $\Delta\Omega$ is the solid angle subtended by each spaxel (1 and 0.04 arcsec^2 for CALIFA and AMUSING observations, respectively) and D is the distance to the galaxy, estimated from the measured redshift. We have checked that our estimates are consistent with the values provided by CALIFA collaboration[34] for the entire galaxy (see Figure 2), and extended this scheme to AMUSING data. As a result of this analysis, we divided our sample into logarithmic mass bins spaced 0.5 dex.

Our morphological classification follows the catalogue provided by the CALIFA collaboration, based on visual inspection by several team members[34]. Regarding data from AMUSING, Hubble type morphology has been recollected from different sources such as that provided by NASA/IPAC Extragalactic Database (NED). We distinguish between elliptical (E), lenticular (S0 and S0a, referred to as $S0$), early- (Sa, Sab, Sb, and Sbc as Sa, Sb) and late-type spirals (Sc, Scd, Sd, Sdm and Sm as Sc, Sd, Sm), without any Irregular galaxy in the sample, and not taking into account Barredness or Interaction State.

Nuclear activity has been categorised according to the relative intensity of different emission lines. More precisely, we adopt the widely used[39;41;40] BPT diagrams employing diagnostic ratios [OIII]/Hβ, [NII]/Hα (whose relation is a good way of measuring the mean energy of an ionizing photon) and [SII]/Hα, based on the the intensities of the collisionally-excited forbidden lines [OIII]λ5007, [NII]λ6584, and [SII]λ6717+6731 relative to that of the Balmer series recombination lines Hα and Hβ.

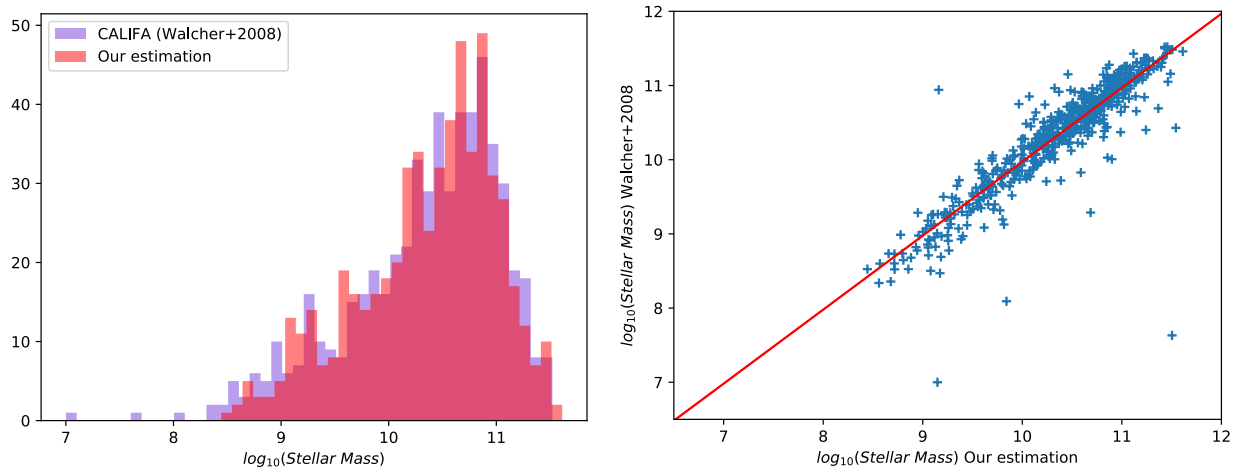

Figure 2: Comparison of our estimate for the galactic stellar mass and that provided by CALIFA collaboration, based on the SDSS ugriz Petrosian magnitudes using code and methods of Walcher et al. 2008[38]. The left graph shows a histogram confronting both methods for each mass value (so as to compare the shape of the distribution), and in the right plot we show the $log_{10}(Stellar\ Mass)$ for each galaxy. We can see that masses are well-correlated (one-to-one guidance is plotted in red), except for those corresponding with the lightest objects in CALIFA classification, whose error could be consider as an anomaly.

Our classification, largely following the criteria outlined in Kewley et al. 2006[40], comprises:

1. **Star-forming galaxies (SFG):** classified as non-active in *either* of the [OIII]/Hβ–[NII]/Hα or [OIII]/Hβ–[SII]/Hα planes. The identification constraints were:

   - $\log([\mathrm{OIII}]/\mathrm{H}\beta) < 0.61/[\log([\mathrm{NII}]/\mathrm{H}\alpha) - 0.05] + 1.3$
     or
   - $\log([\mathrm{OIII}]/\mathrm{H}\beta) < 0.72/[\log([\mathrm{SII}]/\mathrm{H}\alpha) - 0.32] + 1.30$

2. **Composite galaxies (CG):** objects whose spectra may contain significant contributions from both a SMBH as well as star formation (they are likely to contain a young, metal-rich stellar population plus an AGN).

   - $0.61/[\log([\mathrm{NII}]/\mathrm{H}\alpha) - 0.05] + 1.3 < \log([\mathrm{OIII}]/\mathrm{H}\beta)$
     and
   - $\log([\mathrm{OIII}]/\mathrm{H}\beta) < 0.61/[\log([\mathrm{NII}]/\mathrm{H}\alpha) - 0.47] + 1.19$

3. **Low-ionization nuclear emission-line regions (LINER):** these galaxies, whose energy source is still heavily debated, typically display strong lines from neutral or weakly ionized

species (e.g. O, O+, N+, S+) but relatively weak emission from higher ionisation states (e.g. O++, Ne++, He+).

- $0.61/[\log([\mathrm{NII}]/\mathrm{H}\alpha) - 0.47] + 1.19 < \log([\mathrm{OIII}]/\mathrm{H}\beta)$
  and
- $0.72/[\log([\mathrm{SII}]/\mathrm{H}\alpha) - 0.32] + 1.30 < \log([\mathrm{OIII}]/\mathrm{H}\beta)$
  and
- $\log([\mathrm{OIII}]/\mathrm{H}\beta) < 1.89 \log([\mathrm{SII}]/\mathrm{H}\alpha) + 0.76$

4. **Seyfert:** these systems, with the clearest signatures of an AGN, are the low-energy counterpart of the most luminous quasars.

   - $0.61/[\log([\mathrm{NII}]/\mathrm{H}\alpha) - 0.47] + 1.19 < \log([\mathrm{OIII}]/\mathrm{H}\beta)$
     and
   - $0.72/[\log([\mathrm{SII}]/\mathrm{H}\alpha) - 0.32] + 1.30 < \log([\mathrm{OIII}]/\mathrm{H}\beta)$
     and
   - $1.89 \log([\mathrm{SII}]/\mathrm{H}\alpha) + 0.76 < \log([\mathrm{OIII}]/\mathrm{H}\beta)$

5. **Not Classified:** galaxies with one or more negative line intensities (according to the Gaussian fit) that cannot be classified as SFG in neither diagram.

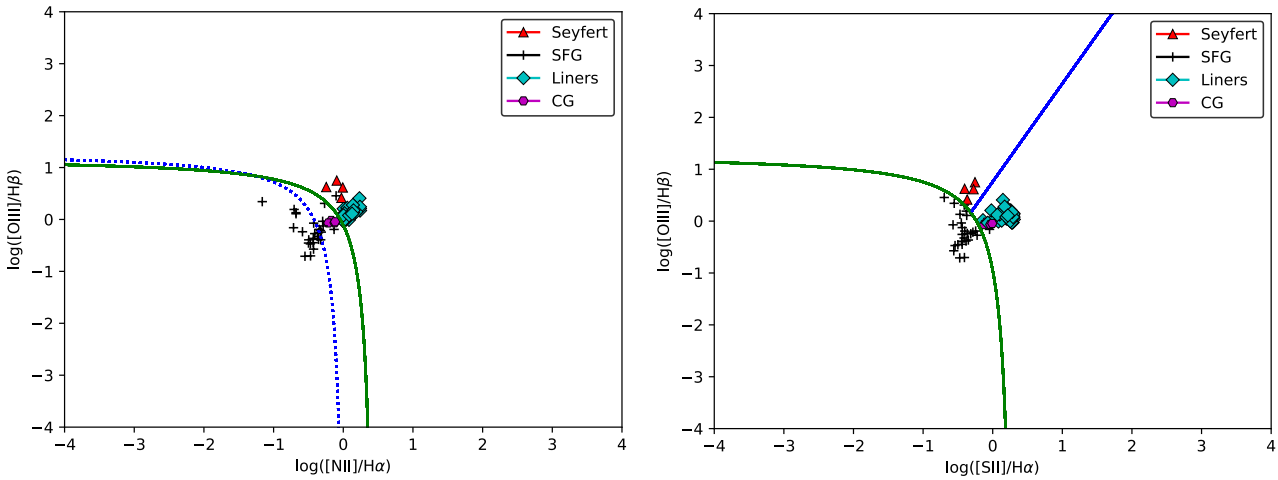

Figure 3: Diagnostic diagrams based on emission-line ratios for AMUSING sample, showing the results based on nuclear (3 pixels circular aperture) spectra. Left panel shows the [OIII]/Hβ–[NII]/Hα plane: measurements above the green solid line[39] require an ionising radiation field harder than the spectrum of an O star (e.g. a SMBH), and most star-forming galaxies are located below the empirical line[41] in dotted blue. The [OIII]/Hβ–[SII]/Hα plane is plotted on the right column: the green solid line[39] is used to separate SFG and AGN, and the blue solid line[40] classifies the latter into Seyfert or LINER galaxies.

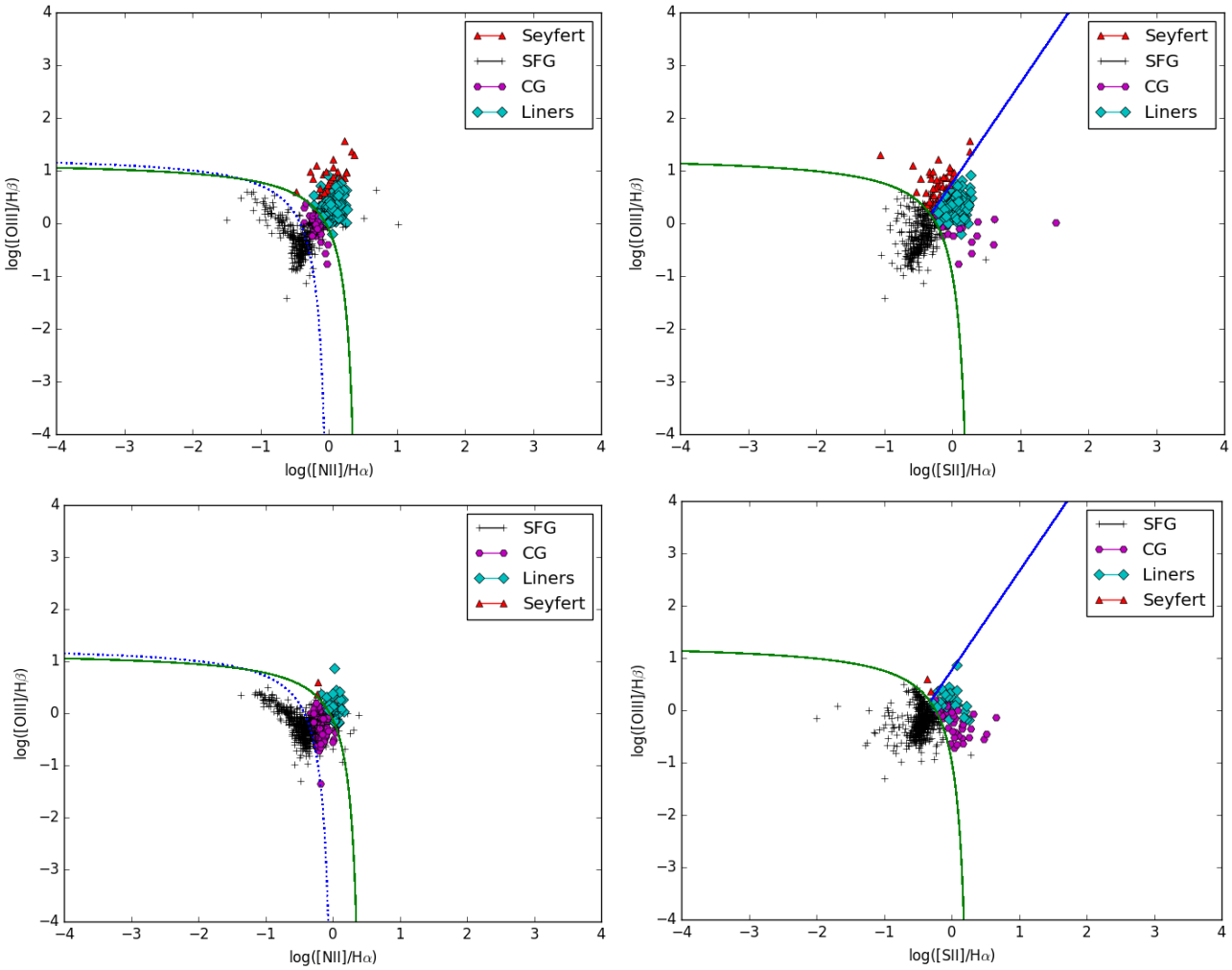

Figure 4: Diagnostic diagrams based on emission-line ratios for CALIFA sample. Left column shows the [OIII]/Hβ–[NII]/Hα plane: measurements above the green solid line[39] require an ionising radiation field harder than the spectrum of an O star (e.g. a SMBH), and most star-forming galaxies are located below the empirical line[41] in dotted blue. The [OIII]/Hβ–[SII]/Hα plane is plotted on the right column: the green solid line[39] is used to separate SFG and AGN, and the blue solid line[40] classifies the latter into Seyfert or LINER galaxies. Top panels show the results based on nuclear (1″ circular aperture) spectra, whereas the bottom panels display the ratios obtained for the entire galaxy in order to illustrate aperture effects that would apply to more distant galaxy samples.

Our CALIFA sample comprises a total of 327 SFG ($\sim$ 56%), 35 CG ($\sim$ 6%), 132 LINERs ($\sim$ 23%), and 47 Seyfert ($\sim$ 8%). 43 galaxies could not be classified. On the other hand, AMUSING sample contains 38 SFG ($\sim$ 36%), 4 CG ($\sim$ 5%), 43 LINERs ($\sim$ 52%), and 4 Seyfert ($\sim$ 5%). 1 galaxy could not be classified. Each of the four categories is plotted with different symbols in the two diagnostic diagrams considered, shown in Figure 3 (AMUSING) and in the top panels of

Figure 4 (CALIFA). The bottom panels (Figure 4) show the result that would be obtained if the integrated spectrum of the entire galaxy was used for classification, as it would occur for a sample of objects at larger redshifts. In such case, the spectral features of the nucleus are diluted due to the larger contribution of the stellar population of the galaxy, and our CALIFA sample would be classified into 529 SFG ($\sim 80\%$), 67 CG ($\sim 10\%$), 36 LINERs ($\sim 5.5\%$), 2 Seyfert ($\sim 0.3\%$), and 18 unclassified spectra (i.e. AGN become substantially more difficult to identify).

3. Reconstruction of the Star Formation History

The stellar mass surface density in every spaxel xy can be derived from the light fractions α^{xy}_{SSP} returned by Pipe3D by combining them with the local surface brightness I^{xy}, the initial mass-to-light ratio Υ_{SSP} of each simple stellar population, and the mass fraction of stars m^{alive}_{SSP} that are still alive at present time:

$$\Sigma^{xy} = 4\pi I^{xy} \sum_{SSP} \alpha^{xy}_{SSP} \Upsilon_{SSP}\, m^{alive}_{SSP} \tag{1}$$

The total mass of stars *formed* over a certain time τ within a given region R is obtained by summing the contribution of SSPs with ages $\tau_{SSP} < \tau$ over all spaxels in region R (each of them representing a physical area $a = \Delta\Omega\ D^2$ in the reference frame of a galaxy located at an angular diameter distance D)

$$M(\tau) = 4\pi a \sum_{xy\in R} I^{xy} \sum_{\tau_{SSP}<\tau} \alpha^{xy}_{SSP} \Upsilon_{SSP} \tag{2}$$

Let us represent the SFH in terms of the average star formation rate $\langle SFR\rangle$ on a time scale τ, normalised to the overall history of the galaxy:

$$\frac{\langle SFR(\tau)\rangle}{\langle SFR(t_0)\rangle} = \frac{\frac{M(0)-M(\tau)}{\tau}}{\frac{M(0)}{t_0}} = \frac{t_0}{\tau}\left[1 - \frac{M(\tau)}{M(t_0)}\right] \tag{3}$$

where $\tau = 0$ corresponds to present time and t_0 is the current age of the universe. If a galaxy, or a region within, had undergone a recent "quenching" episode at $\tau_Q \ll t_0$, the current value of this function would be much less than unity, rising sharply for ages larger than τ_Q.

IV. Results

To analyse the possible effects of an AGN on its environment and the entire galaxy, it has been followed this scheme:

- **Nucleus type and galaxy morphology correlation:** (see Figures 5 and 7) Here we reproduce the median of SFHs and the standard deviation for each galaxy type, regarding

nucleus type and morphology, as well as individual SFHs in grey. In the rightmost column, galaxies with same morphology but different nucleus type are plotted together whereas the bottommost row represents galaxies with same nucleus type for each different morphology.

- **Nucleus type and galaxy mass correlation:** (see Figure 6 and 8) In this case, we reproduce the median of SFHs and the standard deviation for each galaxy, depending on nucleus type and the stellar mass of the entire galaxy. The individual SFHs for each galaxy are plotted in grey. The rightmost column represents galaxies within the same mass range but with different nucleus type, while the bottommost row shows galaxies with same nucleus type for the entire range of masses analysed.

1. Entire galaxy scale (CALIFA sample)

The analysis of the entire galaxy has been reproduced, using the new CALIFA DR3 survey (AMUSING was only used for the circumnuclear analysis due to FoV restrictions), with the aim of having and updated comparative sample. Galaxy morphology has been taken from the catalogue provided by CALIFA, while nucleus type has been computed using the aforementioned emission lines from the nuclear region.

The aim of this analysis is to examine if there exists any difference in the behaviour of SFH among different nucleus types, or if the only correlation appears depending on galaxy morphology.

Figures 5 and 6 show that the behaviour of the SFH for the entire galaxy scale is not dominated by nucleus type. In order to avoid a displacement of the average dominated by the most dispersed individual values (as it happens with a linear average), or a false *global "quenching" behaviour* due to *individual "quenching" behaviour* (result of a geometric mean), we have represented the median of SFHs. In the rightmost column of Figure 5 we find that these median values show a very similar behaviour for all nuclei. However, a stratification as a function of galaxy morphology can be appreciated in the bottommost row, especially in the most populated SFG sample.

A particular point that can be well appreciated in this stratification is that spiral galaxies (specially late-types) display a growing SFH (ie. "they are forming stars"), whilst elliptical galaxies tend to show a decrease followed by stagnation. This pattern agrees with the bimodal behaviour discussed in the introduction, which is also supported by Figure 6, where it is shown that more massive galaxies behave in the same way as ellipticals whereas the lighter ones behave alike spirals.

2. Circumnuclear region (CALIFA and AMUSING sample)

A weak correlation between the SFH for the circumnuclear region with nucleus type (using CALIFA and AMUSING sample together) can be seen in Figure 7. In spite of the persistent

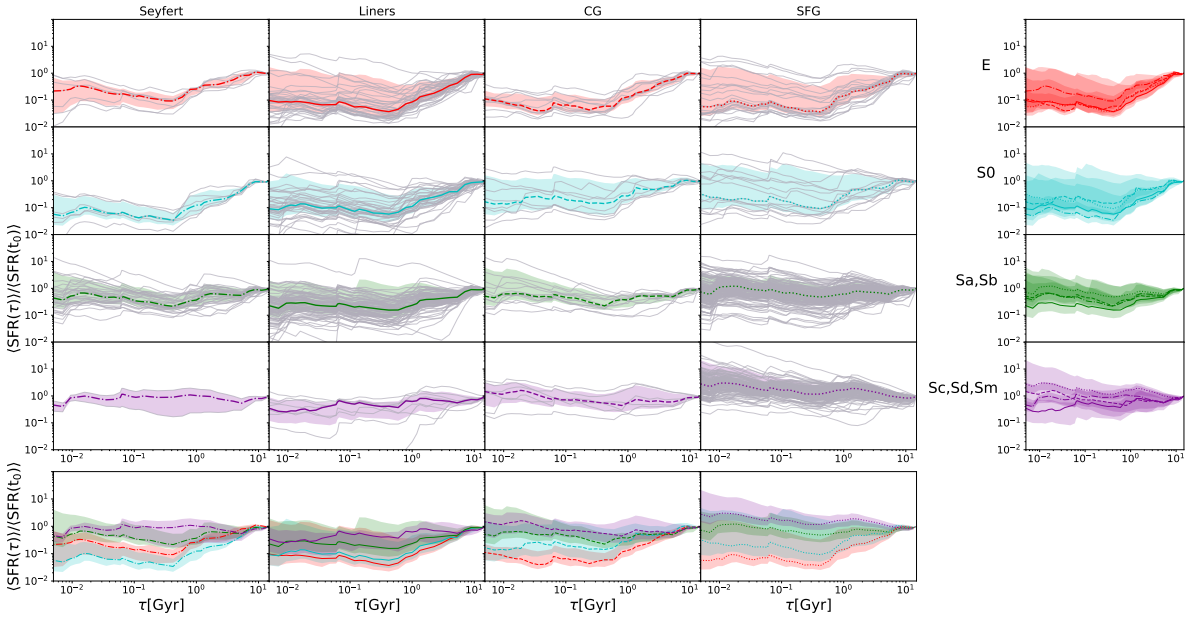

Figure 5: Star Formation History for each galaxy (grey lines) regarding nucleus type (columns) and galaxy morphology (rows). The median of the SFH is represented as coloured lines. The rightmost column confronts the behaviour of the different nuclei for a given morphology, while the bottommost row represents it for a concrete nucleus and all morphology types.

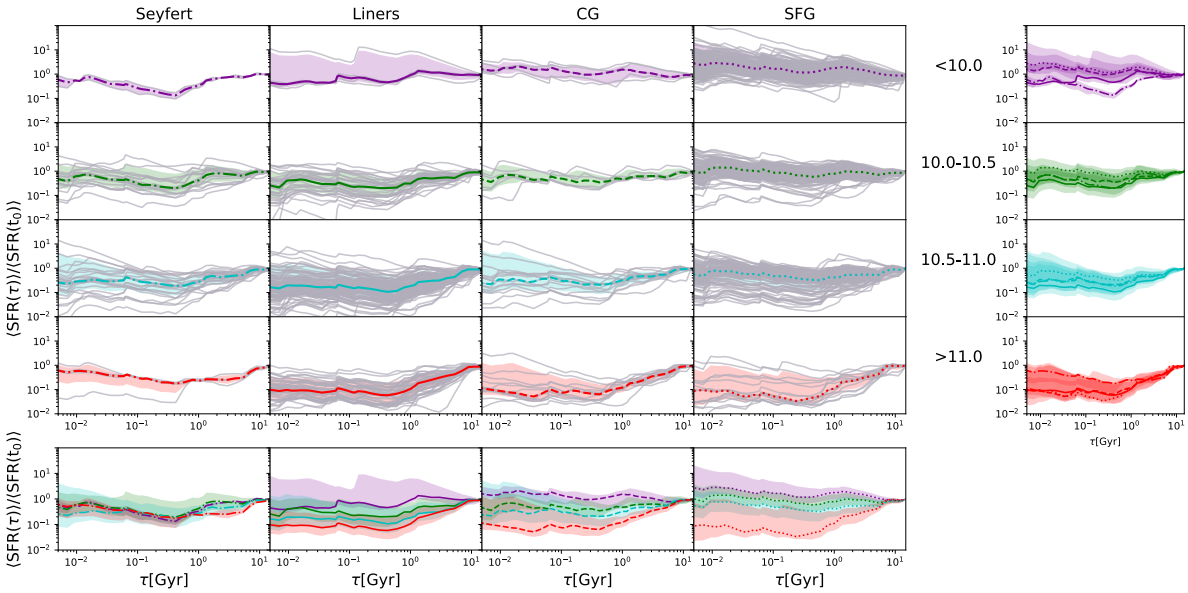

Figure 6: Star Formation History for each galaxy (grey lines) regarding nucleus type (columns) and mass of alive stars (rows). The median of the SFH is represented as coloured lines. The rightmost column confronts the behaviour of the different nuclei for a given mass range, while the bottommost row represents it for a concrete nucleus and all masses.

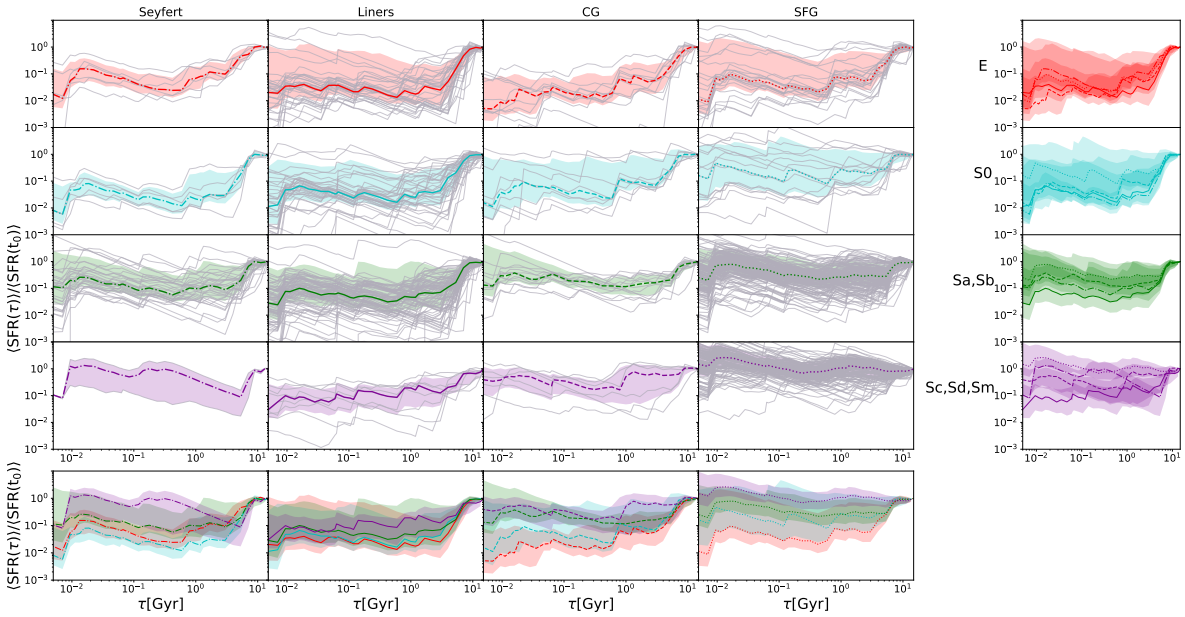

Figure 7: Star Formation History of the circumnuclear region for each galaxy (grey lines) regarding nucleus type (columns) and galaxy morphology (rows). The geometric mean of the SFH is represented as coloured lines. The rightmost column confronts the behaviour of the different nuclei for a given morphology, while the bottommost row represents it for a concrete nucleus and all morphology types.

correlation with morphology (e.g. SFH for elliptical galaxies decreases over time), objects classified as AGN host galaxies have a sharper decrement at early times. This tendency can be appreciated in the rightmost column, where the different medians are more disperse than in Figure 5, and presents different behaviours. In the bottomost, LINERs stand out because of having a stronger correlation, which may have been extended to late-type spiral Seyfert galaxies if they had a bigger representation in our sample.

This behaviour can be extended to Figure 8, where the remaining correlation with stellar mass is weakely modified by the effect of a nuclear AGN, while the SFG sample maintains its behaviour with regards to Figure 6.

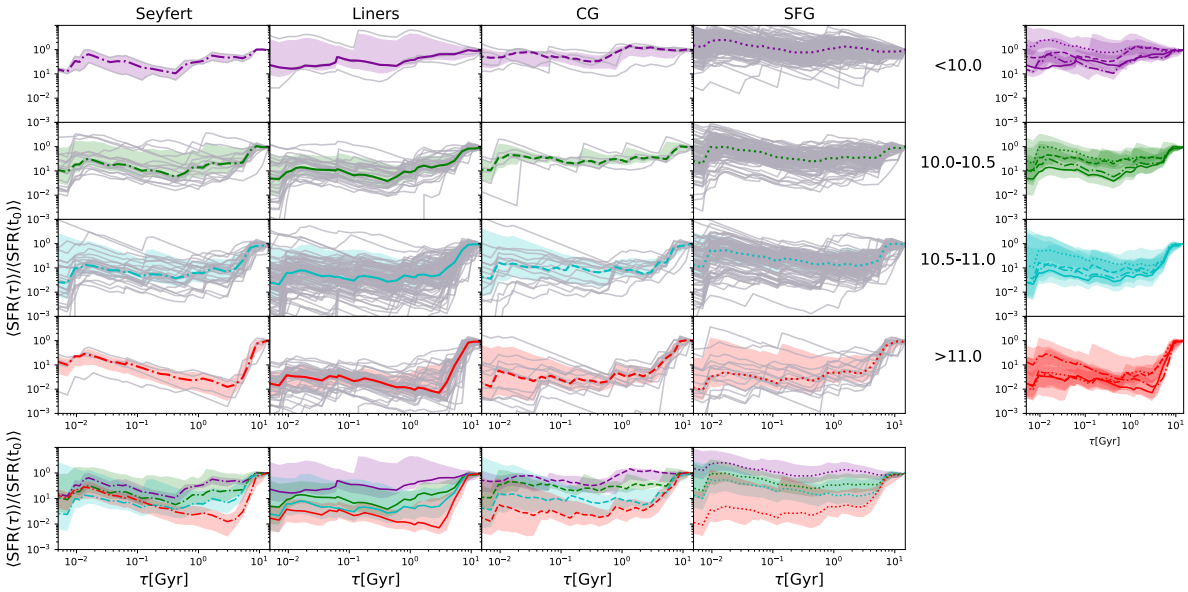

Figure 8: Star Formation History of the circumnuclear region for each galaxy (grey lines) regarding nucleus type (columns) and mass of alive stars (rows). The geometric mean of the SFH is represented as coloured lines. The rightmost column confronts the behaviour of the different nuclei for a given mass range, while the bottommost row represents it for a concrete nucleus and all masses.

V. Conclusions

The analysis carried out using CALIFA and AMUSING samples has allowed the study of the effects that an AGN can cause on its environment.

Firstly, as obtained by a previous work[29], the SFH at galaxy scale seems unaffected by the presence of different nuclei and displays a clear correlation with morphology and mass. These results seem to deny one of the current paradigms, where the AGN would be the source of the transition from the "blue cloud" towards the "red sequence", and still leaves open why so many AGN are found in the "green valley" and whether it has any causal effect on the evolution.

On the other hand, this study has focused mainly on the analysis of circumnuclear regions, trying to find if the presence of an AGN can at least modify the star formation of its surroundings. We have observed a weak correlation in the LINER's case, but our conclusions about Seyfert galaxies are strongly limited by their scarceness in our sample. In the LINER case, galaxies seem to have formed the vast majority of their stars in times very close to their birth, with the formation rate sharply decreasing at about 5Gyr, where we observe a departure from the SFG behaviour. In any case, further observations in X-ray frequencies and other wavelengths would be needed to assure the presence of an AGN at their centres. Hence, we are led to the following question: are these galaxies classified as LINERs owing to the presence of an AGN at their centre that reduces

their SFR, or is it the case that the observed line ratios (and the subsequent classification as AGNs in the BPT diagram) are just due to the lack of star formation.

Concerning the "quenching" behaviour, it has only been observed in isolated cases in the CALIFA sample around 1Gyr ago. On the other hand, AMUSING data present much clearer instances of this phenomenon, but owing to the fact that MUSE does not cover the 4000Å break (which contains important parameters used by PIPE3D), the accuracy of our conclusions for stellar populations (e.g. their SFH) will be probably affected[32]. This problem will not be solved until newer methods to treat MUSE data and complementary observations are available.

Summarising, we have not seen any relevant influence of the nucleus on its host galaxy at global scales. Regarding the circumnuclear region, there is a correlation between the nucleus type and the SFH, with AGN-candidates and SFG showing some different trends, but these data can not be taken as definitive.

VI. Acknowledgements

This work would not have been possible without the dedication and guidance of Dr. Y. Ascasibar, who has always been close even when being on the other side of the Earth. I would also like to thank the group of people who week after week attended the extra-galactic group meetings for their ideas and support. I really enjoyed working with you all!.

An important mention in this work should also go to Dr. S. F. Sánchez, for providing us with all the necessary data from CALIFA and AMUSING surveys for this study.

This research has made use of the NASA/IPAC Extragalactic Database (NED) which is operated by the Jet Propulsion Laboratory, California Institute of Technology, under contract with the National Aeronautics and Space Administration. There has also been use of data provided by the Calar Alto Legacy Integral Field Area (CALIFA) survey, based on observations collected at the Centro Astronómico Hispano Alemán (CAHA) at Calar Alto, operated jointly by the Max-Planck-Institut für Astronomie and the Instituto de Astrofísica de Andalucía (CSIC).

Last but not least, I would like to thank my family for their support and kindness, my friends, specially Álvaro Muñoz and Miguel Sacristán, always willing to help me when I almost destroyed my computer, and of course, Gabriel Larios, for his feedback and patience, even these last days when I became a bit neurotic.

References

[1] I. Strateva et al., *Color Separation of Galaxy Types in the Sloan Digital Sky Survey Imaging Data*, The Astronomical Journal **122**, 1861 (2001).

[2] I. K. Baldry et al., *Quantifying the bimodal color-magnitude distribution of galaxies*, The Astronomical Journal **600**, 681 (2004).

[3] I. K. Baldry et al., *Galaxy bimodality versus stellar mass and environment*, MNRAS **373**, 469 (2006).

[4] E. F. Bell et al., *Nearly 5000 distant early-type galaxies in COMBO-17: A red sequence and its evolution since z∼1*, The Astrophysical Journal **608**, 752-767 (2004)

[5] S. M. Faber et al., *Galaxy luminosity functions to z∼1 from DEEP2 and COMBO-17: implications for red galaxy formation*, The Astrophysical Journal **665**, 265-294 (2007)

[6] T. K. Wyder et al., *The UV-optical color magnitude diagram. I. Basic properties* **173**, 293-314 (2007)

[7] D. Schiminovich et al., *The UV-optical color magnitude diagram. III. Physical properties and morphological evolution on and off of a star-forming sequence*, The Astrophysical Journal Supplement Series **173**, 315-341 (2007)

[8] D. C. Martin et al., *The UV-optical galaxy color-magnitude diagram. III. Constraints on evolution from the blue to the red sequence*, The Astrophysical Journal Supplement Series **173**, 342-356 (2007)

[9] A. J. Mendez et al., *AEGIS: The morphologies of green galaxies at 0.4 <z< 1.2*, The Astrophysical Journal **736**, 110 (2011)

[10] T. S. Gonçalves et al., *quenching star formation at intermediate redshifts: Downsizing of the mass flux density in the green valley*, The Astrophysical Journal **759**, 67 (2012)

[11] S. Salim et al., *UV star formation rates in the local universe*, The Astrophysical Journal Supplement Series **173**, 267-292 (2007)

[12] K. Schawinski et al., *Observational evidence for AGN feedback in early-type galaxies*, MNRAS **382**, 1415-1431 (2007)

[13] K. Schawinski et al., *The role of mergers in early-type galaxy evolution and black hole growth*, The Astrophysical Journal Letters **714**, L108-L112 (2010)

[14] E. A. Fath, *The spectral of some spiral nebulae and globular stars clusters*, Lick Observatory Bulletin **149**, 71-77 (1909)

[15] C. K. Seyfert, *Nuclear emission in spiral nebulae*, Astrophysical Journal **97**, 28 (1943)

[16] C. Flynn, *Lectures on advanced Astrophysics*, **6** (2005)

[17] J. Magorrian et al., *The demography of massive dark objects in galaxy centers*, The Astronomical Journal **115**, 2285-2305 (1998)

[18] L. Ferrarese, D. Merrit, *A fundamental relation between supermassive black holes and their host galaxies*, The Astrophysical Journal **539**, L9-L12 (2000)

[19] K., Gebhardt et al., *A relationship between nuclear black hole mass and galaxy velocity dispersion*,The Astrophysical Journal **539**, L3-L16 (2000)

[20] R. Ganguly, M. S. Brotherton, *On the fraction of quasars with outflows*, The Astrophysical Journal **672**, 102-107 (2008)

[21] F. Tombesi et al., *Discovery of Ultra-Fast Outflows in a sample of broad-line radio galaxies observed with SUZAKU*, The Astrophysical Journal **719**, 700-715 (2010)

[22] J. Silk, M. J. Rees, *Quasars and galaxy formation*, Astronomy and Astrophysics **331**, L1-L4 (1998)

[23] T. Di Matteo et al., *Energy input from quasars regulates the growth and activity of black holes and their host galaxies*, Nature **433**, 604-607 (2005)

[24] N. Menci et al., *The abundance of distant and extremely red galaxies: the role of AGN feedback in hierarchical models*, The Astrophysical Journal **647**, 753-762 (2006)

[25] A. C. Fabian, *Observational Evidence of Active Galactic Nuclei Feedback*, Annual Review of Astronomy and Astrophysics **50**, 455 (2012)

[26] K. Zubovas, A. King, *Clearing out a galaxy*, The Astrophysical Journal Letters **745**, L34 (2012)

[27] Y. Ascasibar, J. Sánchez Almeida, *Do galaxies form a spectroscopic sequence?*, MNRAS **415**, 2417-2425 (2011)

[28] J. Casado et al., *Nature or nurture? Clues from the distribution of specific star formation rates in SDSS galaxies*, MNRAS **451(1)**, 888-903 (2015)

[29] N.P. Plaza, *Impacto de los núcleos galácticos activos sobre la formación estelar a escala galáctica*, Trabajo de fin de Grado - Universidad Autónoma de Madrid (2016).

[30] K. Bundy et al., *Overview of the SDSS-IV MaNGA survey: mapping nearby galaxies at Apache Point Observatory*, The Astrophysical Journal **798**, 7 (2015)

[31] S. M. Croom et al., *The Sydney-AAO Multi-object Integral field spectrograph*, MNRAS **421**, 872-893 (2012)

[32] S. F. Sánchez et al., *CALIFA, the Calar Alto Legacy Integral Field Area survey IV. Third public data release*, Astronomy & Astrophysics **594**, A36 (2016)

[33] S. F. Sánchez et al., *CALIFA, the Calar Alto Legacy Integral Field Area survey I. Survey presentation*, Astronomy & Astrophysics **538**, A8 (2012)

[34] C. J. Walcher et al., *CALIFA: a diameter-selected sample for an integral field spectroscopy galaxy survey*, Astronomy & Astrophysics **569**, A1 (2014)

[35] S. F. Sánchez et al., *PIPE3D, a pipeline to analyze Integral Field Spectroscopy data: I. New fitting philosophy of FIT3D* , Revista Mexicana de Astronomía y Astrofísica **52**, 21-53 (2016)

[36] S. F. Sánchez et al., *PIPE3D, a pipeline to analyze Integral Field Spectroscopy data: II. Analysis sequence and CALIFA dataproducts* , Revista Mexicana de Astronomía y Astrofísica **52**, 171-220 (2016)

[37] C. J. Walcher et al., *CALIFA: a diameter-selected sample for an integral field spectroscopy galaxy survey*, Astronomy & Astrophysics **569**, A1 (2014)

[38] C. J. Walcher et al., *The VVDS-SWIRE-GALEX-CFHTLS surveys: physical properties of galaxies at z below 1.2 from photometric data*, Astronomy & Astrophysics **491**, 713-730 (2008)

[39] L. J. Kewley et al., *Theoretical modeling of Starburst Galaxies*, The Astrophysical Journal **556**, 121-140 (2001)

[40] L. J. Kewley et al., *The host galaxies and classification of active galactic nuclei*, MNRAS **372**, 961-976 (2006)

[41] G. Kauffmann et al., *The host galaxies of active galactic nuclei*, MNRAS **346**, 1055-1077 (2003)

[42] R. Cid Fernandes et al., *Resolving galaxies in time and space I. Applying STARLIGHT to CALIFA datacubes*, Astronomy & Astrophysics **557**, A86 (2013)

[43] L. P. Martins et al., *A high-resolution stellar library for evolutionary population synthesis*, MNRAS **358**, 49-65 (2005)

[44] S. F. Sánchez, *Techniques for reducing fiber-fed and integral-field spectroscopy data:An implementation on R3D*, Astronomische Nachrichten **327(9)**, 850-861 (2006)

[45] A. Vazdekis et al., *Evolutionary stellar population synthesis with MILES ? I. The base models and a new line index system*, MNRAS **404**, 1639-1671 (2010)

[46] J. Falcón -Barroso et al., *An updated MILES stellar library and stellar population models*, Astronomy & Astrophysics **532**, A95 (2011)

[47] E. E. Salpeter, *The luminosity function and stellar evolution*, Astrophysical Journal **121**, 161 (1995)

[48] R. M. González Delgado et al., *The CALIFA survey across the Hubble sequence Spatially resolved stellar population properties in galaxies*, Astronomy & Astrophysics **581**, A103 (2015)